BUSINESS CASE

LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

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1. Executive Summary

This business case outlines how the LitPath AI project will address the research discovery limitations of the DOST-STII Library's current Online Public Access Catalog (OPAC) system, the anticipated benefits of the project, and the recommendation and justification for moving forward. The business case also discusses detailed project goals, performance measures, assumptions, constraints, and alternative options considered.

The DOST-STII Library maintains a valuable collection of graduate theses and dissertations, yet this collection remains significantly underutilized due to the limitations of its current OPAC, specifically its reliance on exact keyword matching, lack of AI-powered search, slow performance, and missing citation and bookmarking tools. These limitations create barriers for student researchers, DOST employees, and academic professionals who depend on this collection for their work.

LitPath AI is proposed as a standalone AI-powered web application designed to complement the existing OPAC. Rather than modifying or replacing the current system, it provides an independent platform that enhances research discovery through advanced AI capabilities. By leveraging semantic search, automated citation generation, bookmarking features, and analytics, LitPath AI significantly improves research efficiency and accessibility for users.

1.1. Issue

The DOST-STII Library's current Online Public Access Catalog (OPAC) is limited in its ability to support modern research needs. The system relies solely on exact keyword matching, making it difficult for users who are exploring new topics or are unsure of precise terminology to discover relevant materials. Internal assessments show that fewer than 7 out of 10 search results align with the user's actual research interests, and users must spend approximately 15 minutes manually sifting through results to find useful materials. Search queries also take over a minute to return results which is far exceeding the acceptable standard, and the system lacks essential features such as built-in citation and bookmarking tools. These deficiencies result in research inefficiencies and the underutilization of the library's key academic resources.

1.2. Anticipated Outcomes

Implementing LitPath AI as a standalone web application for the DOST-STII Library will enhance the research experience, particularly for users searching for theses and dissertations. By augmenting the existing keyword-dependent system with an AI-powered search capability, users will be able to discover relevant academic works even when they are unsure of the exact terminology. This improvement will reduce reliance on precise keyword matching and allow for more intuitive and efficient research.

A built-in automated citation generator supporting APA, MLA, Chicago, and IEEE formats will eliminate the need for users to manually format references, improving accuracy and saving time. In addition, a bookmarking feature will allow users to save and easily revisit selected theses and dissertations, supporting better organization of research materials. The system will also process search queries in under one second, replacing the current one-minute-plus delays with a fast and responsive experience.

For DOST-STII as an institution, LitPath AI will improve service delivery by enhancing access to its academic collections. Librarians will benefit from an admin dashboard that provides analytics on user search trends related to theses and dissertations, enabling more informed decisions in collection development and management.

1.3. Recommendation

Various options and alternatives were analyzed to determine the best way to address the research discovery limitations of the DOST-STII OPAC. The recommended approach is developing LitPath AI as a standalone AI-powered web application for theses and dissertations built by HecTech which was selected because it directly addresses the identified problems, is technically feasible using currently available tools, and does not require a formal budget from the HecTech side, as cloud and AI/LLM operational costs will be shouldered by DOST-STII.

LitPath AI will achieve its desired results through the following:

- AI-powered query analysis that enables the system to understand user research intent beyond exact keyword matches, returning semantically relevant theses and dissertations.
- A built-in citation generator that eliminates the manual effort required to format references in APA, MLA, Chicago, and IEEE styles.
- A bookmarking feature that allows users to save, organize, and quickly access selected research materials for future use.
- An admin dashboard that provides DOST-STII librarians with visibility into search trends and usage patterns related to academic works.

1.4. Justification

LitPath AI was selected as the recommended solution because it directly addresses the critical limitations of the DOST-STII Library's current OPAC while remaining technically and financially feasible for all parties involved. The system was chosen over all other alternatives as it delivers the greatest improvement in research efficiency without requiring a formal development budget from HecTech, since cloud hosting and AI/LLM API operational costs will be covered by DOST-STII.

Key stakeholders including Science Research Specialists Khasian Eunice Romulo and Jonathan Abalon have confirmed the need for modernized search capabilities and improved user satisfaction with the library's thesis and dissertation collection. Their direct involvement in requirements definition and deliverable approval further validates the alignment of LitPath AI with DOST-STII's institutional objectives.

If LitPath AI is not implemented, the DOST-STII Library will continue to operate with a search system that fails to meet modern research expectations. Users will continue experiencing poor search relevance, slow query performance of over 2–3 minutes, and the absence of essential tools such as citation generation and bookmarking, directly impacting the productivity of students, academic researchers, and DOST employees who rely on the library's collection.

Initial estimates for the LitPath AI project are:

- At least 70% of search relevance accuracy in the top 10 results, based on user feedback, compared to the current rate of fewer than 7 out of 10 relevant results
- Search queries are processed and returned significantly faster compared to the existing system, resulting in more responsive and efficient user experience.
- Elimination of manual citation formatting through an automated citation generator supporting APA 7th, MLA 9th, Chicago, and IEEE formats
- Reduction in time spent manually filtering through irrelevant results from approximately 15 minutes per session to just a few minutes through improved semantic search relevance

2. Business Case Analysis Team

The following individuals comprise the business case analysis team. They are responsible for the analysis and creation of the LitPath AI project business case.

Role	Description	Name/Title
Project Manager	Manages business case development and leads the project team	Tracie Tomon, Group Leader, HecTech
Project Adviser	Provides project guidance	Gonzalo Guumogda, SoCIT Faculty
Development Team	Contributes to business case analysis through system development context and technical inputs	HecTech Team, Project Developers
Client	Provides project requirements and approval	Khasian Romulo, Science Research Specialist II

3. Problem Definition

3.1. Problem Statement

The DOST-STII Library houses a valuable and growing collection of graduate theses and dissertations from DOST Scholars, cataloged through its Online Public Access Catalog (OPAC). Despite the value of this collection, it remains underutilized due to critical limitations in the OPAC's search functionality, including exact keyword matching, slow processing times, and a lack of Artificial Intelligence. These limitations create a significant barrier to knowledge discovery for students, academic researchers, and DOST employees. Users who are exploring new research areas or are unsure of precise terminology find it difficult to discover relevant materials, leading to research inefficiencies and preventing the library from maximizing the impact of its key academic resources.

Specifically, the key challenges with the current system are:

1. Limited Search Mechanism: The current system relies heavily on exact keyword and title-based searches, lacking AI-powered search capabilities to interpret a user's research interests or intent. This makes the system ineffective for users exploring broader topics or those unfamiliar with precise terminology, often yielding results that are irrelevant or incomplete.
2. Poor Relevance and Efficiency: The OPAC struggles to consistently generate accurate results due to the absence of advanced, intent-driven search algorithms. Internal assessments reveal that fewer than 7 out of 10 results match a user's actual research interests. Consequently, users are forced to spend approximately 15 minutes per session manually reviewing and sifting through irrelevant results to find useful materials.
3. Lengthy Searching Performance: The search function exhibits severe performance issues, with search queries taking over a minute to return results. This processing time is significantly slower than the acceptable industry standard of 30 seconds, leading to a poor user experience and discouraging further exploration of the platform.
4. Lack of Essential Research Support Features: The current system lacks key built-in tools essential for modern academic research, specifically automated citation generation and bookmarking capabilities. As a result, users must manually compile citations and manage their research materials independently, which increases the time spent on administrative tasks and raises the risk of formatting errors or disorganized references.

3.2. Organizational Impact

LitPath AI will affect DOST-STII and the HecTech development team in the following ways:

Tools:
The existing OPAC will remain in place as a separate system, while LitPath AI operates as a standalone web application focused on theses and dissertations. Users can access LitPath AI independently to take advantage of its advanced AI-powered search capabilities without modifying the current OPAC. This approach ensures no disruption to existing operations while providing a significantly enhanced research experience. In addition, DOST-STII staff will be provided with an admin dashboard to monitor usage analytics and gain insights into user behavior.

Processes:
Research workflows for students, researchers, and DOST employees will be streamlined. Users will be able to quickly discover relevant theses and dissertations, generate citations automatically, and bookmark selected materials for future reference. This reduces the need to manually sift through irrelevant results or manage references separately.

Roles and Responsibilities:
DOST-STII library staff will take on a new role in monitoring and interpreting analytics from the admin dashboard. HecTech will continue to support the system during testing and deployment phases, with full technical documentation and a user manual provided upon project completion.

Hardware/Software:
The system is currently deployed on a Proxmox server instance hosted on a team member's local device for development and testing purposes. For final production deployment, the system will be transitioned to a hosting environment to be confirmed with DOST-STII, with associated costs shouldered by the client. AI/LLM API services will also be covered by DOST-STII. No additional hardware purchases are required from HecTech

3.3. Technology Migration

LitPath AI will be implemented using a phased approach to ensure continuity of the existing OPAC system and minimize disruption to DOST-STII operations.

- Phase I — Planning & Design (Apr–May 2025, Completed):

  - Project scoping, requirements gathering, stakeholder interviews, and UI/UX design using Figma.
- Phase II — Core Development (Sep 2025–Jan 2026, Completed):
  - Development of the AI search engine, core UI, citation generator, bookmarking feature, feedback mechanism, and admin dashboard.
- Phase III — Feature Refinement & Client Integration (Feb–Mar 2026, Completed):
  - Integration of client feedback, continued feature development (research history, follow-up questions), and performance optimization.
- Phase IV — Internal Testing & Temporary Deployment (Apr 2026):
  - Deployment on Proxmox server for internal testing, unit testing, and comprehensive testing of the standalone system prior to final production handover.
- Phase V — Usability Testing & Bug Fixes (May–Jun 2026):
  - Usability testing with student researchers and DOST employees, performance optimization, and bug fixes.
- Phase VI — Final Deployment (Jun–Jul 2026):
  - Full production deployment as a standalone system, operating independently from the existing DOST-STII OPAC, followed by the formal handover of the system and all documentation.

4. Project Overview

The LitPath AI Project Overview provides detail for how this project will address the DOST-STII Library's research discovery problem. The overview covers the project description, goals and objectives, performance criteria, assumptions, constraints, and major milestones.

4.1. Project Description

LitPath AI is a standalone AI-powered web application developed by HecTech for the DOST-STII Library. Operating independently from the existing OPAC, the system enhances the discovery of graduate theses and dissertations by moving beyond simple keyword matching. It understands user research intent through AI-powered query analysis, returning semantically relevant results.

In addition to improved search functionality, the system includes a built-in citation generator and a bookmarking feature, enabling users to efficiently manage and organize their research materials.

The project will be completed through six development and deployment phases, leveraging the four-member HecTech team's expertise in software development, AI integration, and UI/UX design. Upon final deployment in June–July 2026, DOST-STII will receive a fully operational system, along with complete technical documentation and a user

manual. All cloud hosting and AI/LLM API costs will be covered by DOST-STII as the project's primary client and sponsor.

4.2. Goals and Objectives

The LitPath AI project directly supports the research and information management goals of DOST-STII. The following table lists the business goals and objectives that the LitPath AI project supports and how it addresses them:

Business Goal/Objective	Description
Improve research discovery and search relevance	LitPath AI replaces exact keyword matching with AI-powered semantic search using RAG and Gemini AI, enabling users to find relevant theses and dissertations even without precise terminology, targeting at least 70% relevance accuracy in top 10 results
Enhance user efficiency and reduce manual workload	Automated citation generation in APA 7th, MLA 9th, Chicago, and IEEE formats, along with bookmarking features, eliminates manual formatting tasks and reduces time spent managing research materials from approximately 15 minutes to just a few minutes per session
Improve system performance and user experience	Search queries are processed and returned significantly faster compared to the existing system, resulting in more responsive and efficient user experience.
Support informed collection management for DOST-STII	An admin analytics dashboard will provide DOST-STII librarians with visibility into search trends and usage patterns to guide collection development and institutional decision-making

4.3. Project Performance

The following table lists the key resources, processes, or services and their anticipated business outcomes in measuring the performance of the project. These performance measures will be quantified and further defined in the detailed project plan.

Key Resource/Process/Service	Performance Measure
Search Relevance	Top 10 search results achieve a minimum of 70% relevance accuracy based on user feedback and expert validation
User Satisfaction	Average satisfaction score of at least 3.5 out of 5 from usability testing participants
System Response Time	Search queries processed and results displayed faster compared to the existing system
Admin Dashboard	DOST-STII librarians able to access and interpret search trend data to inform collection management

4.4. Project Assumptions

This section outlines the key conditions considered true for the planning and scoping of the LitPath AI project. If any of these assumptions prove false, it may introduce risks that impact on the timeline, budget, or scope, requiring a formal change request:

- The current library servers or the planned cloud hosting environment has the capacity to support the AI engine
- The project team will have continuous and stable access to the DOST-STII database and necessary technical systems throughout the project timeline.
- The project scope will remain stable throughout the development phase.
- The Project Sponsors, Khasian Eunice Romulo and Jonathan Abalon, will remain available for milestone approvals, client feedback, and formal acceptance of deliverables.
- Cloud hosting and AI/LLM API costs will be fully covered by DOST-STII as agreed.
- Users attempting to log in to LitPath AI will be able to register directly in the system, independent of the parent OPAC.

4.5. Project Constraints

This section outlines the specific limitations and boundaries within which the LitPath AI project must be executed. This strict schedule, resource, financial, and scope parameters dictate the team's operations and must be adhered to throughout the project lifecycle:

- The project must be fully deployed by June–July 2026 in alignment with the academic and institutional timeline.
- Development is limited to the four-member HecTech team with no external outsourcing anticipated.
- No formal development budget is allocated to HecTech, and the team's primary resource is time and effort.
- Full-text access to undigitized physical theses/dissertations is out of scope; the system focuses on abstracts and metadata only.

4.6. Major Project Milestones

The following are the major project milestones identified at this time. As the project planning moves forward and the schedule is developed, the milestones and their target completion dates will be modified, adjusted, and finalized as necessary to establish the baseline schedule.

Milestones/Deliverables	Target Month
Project Scoping	April 2025

Information Gathering	April – May 2025
Completion of Project Deliverables & Documentation	June 2025
Design Thinking Process (Stages 1–5)	August 2025
Core UI Development	September – October 2025
AI Search Engine, Feedback Mechanism & Admin Dashboard Development	October 2025 – January 2026
Feature Refinement & Client Feedback Integration	February – March 2026
Continued Development & Performance Optimization	March – April 2026
Proxmox Deployment & Internal Testing	April 2026
Bug Fixes & Usability Testing	May – June 2026
Final Deployment & Project Handover	June 2026

5. Strategic Alignment

The LitPath AI project is in direct support of DOST-STII's institutional goals and HecTech's academic development objectives. By supporting these strategic plans, this project advances knowledge accessibility and research innovation within the institution.

Plan	Goals/Objectives	Relationship to Project
DOST-STII Digital Information Access Plan	Improve accessibility and discoverability of academic resources, particularly graduate-level theses and dissertations, for researchers, students, and DOST employees	LitPath AI directly enhances the discoverability of the DOST-STII Library's thesis and dissertation collection through AI-powered semantic search using RAG and Gemini AI, making the collection significantly more accessible and useful to its target users
DOST-STII Technology Modernization Plan	Leverage modern technology to improve institutional service delivery without disrupting existing systems	LitPath AI operates as an independent, standalone application alongside the existing OPAC system. It introduces AI-powered search, automated citation generation in APA, MLA, Chicago, and IEEE formats, and an admin analytics dashboard, serving as a dedicated research tool without replacing or altering the legacy catalog.
DOST-STII Research Support and Scholar Empowerment Plan	Support DOST scholars and employees in conducting efficient, accurate, and well-documented research	LitPath AI reduces research friction by cutting response time from over 10 minutes to under 5 minutes, delivering semantically relevant results with at least

		70% accuracy, and providing tools that help users organize, bookmark, and cite their materials with ease

6. Cost Benefit Analysis

The following table captures the cost and benefit actions associated with the LitPath AI project, descriptions of these actions, and their estimated costs or savings.

Action	Action Type	Description	First year costs (- indicate anticipated savings)
System Development by HecTech	Cost	Labor and time investment by the 4-member HecTech team across six development phases spanning the 2025–2026 academic year	₱0 (in-kind contribution — no formal budget allocated to HecTech)
Cloud Hosting & Infrastructure	Cost	Currently hosted on a Proxmox server instance on a team member's local device for development and testing. Final production hosting environment and costs to be confirmed with DOST-STII upon deployment.	Shouldered by DOST-STII
AI / LLM API Integration (Gemini AI)	Cost	Operational costs for AI-powered query analysis using Retrieval-Augmented Generation (RAG) and Gemini AI during development, testing, and live operation	Shouldered by DOST-STII
Reduction in manual research time	Benefit	Users currently spend approximately 15 minutes per session manually sifting through irrelevant results. Improved semantic search is estimated to reduce this to just a few minutes per session	Significant time savings per user session upon deployment
Elimination of manual citation formatting	Benefit	Automated citation generation in APA 7th, MLA 9th, Chicago, and IEEE formats removes the need for users to manually format references, reducing errors and saving time per research session	Estimated hours saved per researcher per project
Improved system response time	Benefit	Replacing the current 10-minute query response time with under 5-minute results drastically reduces researcher wait time and improves overall productivity	Immediate upon deployment

Since HecTech contributes its full development effort at no formal cost and DOST-STII shoulders all cloud and AI/LLM operational expenses, the LitPath AI project represents a high-value, low-financial-risk initiative for both parties. The primary return on investment is measured in improved research efficiency, faster search performance, and greater utilization of the DOST-STII Library's thesis and dissertation collection.

7. Alternatives Analysis

The following alternative options have been considered to address the business problem. These alternatives were not selected for several reasons which are also explained below.

No Project (Status Quo)	**Reasons For Not Selecting Alternative**
Keep the existing DOST-STII OPAC system without any enhancements or modifications	• Users continue to experience poor search relevance with fewer than 7 out of 10 results matching their actual research needs • Search queries continue to take over 10 minutes to return results, far exceeding the acceptable industry standard • No citation generation or bookmarking tools available, forcing users to manually compile and format references • The library's thesis and dissertation collection continues to be significantly underutilized due to the limitations of the current search system
Alternative Option	**Reasons For Not Selecting Alternative**
Purchase and deploy a third-party AI-powered library or academic search system	• Significantly higher licensing, subscription, and implementation costs that are outside the project's budget constraints • Third-party solutions are not tailored to the specific structure and content of the DOST-STII OPAC and its thesis and dissertation collection • May require full replacement of the existing OPAC, causing major disruption to ongoing library operations • Vendor solutions may not comply with the Data Privacy Act of 2012 (RA 10173) or DOST-STII's internal security requirements
Alternative Option	**Reasons For Not Selecting Alternative**
Full replacement and redevelopment of the OPAC system from scratch	• Requires a substantially larger budget and a significantly longer development timeline beyond what HecTech can deliver within the 2025–2026 academic year • Poses high operational risk by completely replacing the existing catalog system that users and library staff currently depend on daily • A full redevelopment goes far beyond the defined project scope and the available resources of the five-member HecTech team • Does not align with DOST-STII's preference for a non-disruptive supplementary solution that works alongside the existing OPAC

8. APPROVALS

The signatures of the people below indicate an understanding of the purpose and content of this document by those signing it. By signing this document, you indicate that you approve of the proposed project outlined in this business case and that the next steps may be taken to create a formal project in accordance with the details outlined herein.

Approver Name	Title	Signature	Date
Khasian Eunice Romulo	Science Research Specialist II / Unit Head, Digital Information and Processing Delivery Unit, DOST-STII		